

Lexical creativity in modern Nahuatl: An analysis of multidialectal data

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Abstract

Nahuatl has a long history of contact with, and marginalization by, Spanish. Today the language shift maintains a steady pace, despite numerous initiatives for revitalization and revalorization. This paper deals with the lexical creativity of Nahuatl speakers. As a result of the fieldwork of our research team in six communities in Mexico, we have amassed rich data concerning neologisms that designate objects and concepts imported from European culture. Unlike most studies on language contact in Nahuatl, we choose to focus not on the morphosyntactic strategies of lexicogenesis, or on phenomena of matter and/or pattern borrowing, but on the conceptual structure of neologisms, applying the onomasiological theory of word-formation. We argue that, while language endangerment and the lack of an obvious need for lexicogenesis may cause some productivity patterns to go dormant, these do not necessarily become eradicated. Moreover, innovative patterns of word-formation may come to light when lexical creativity is triggered.

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Lexicogenesis =>

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1. INTRODUCTION

Neology, the study of newly created lexemes, has a somewhat peripheral status in linguistics, even though it represents a key aspect of language evolution – in effect, every word must first be created in order to be pronounced. Lexical expansion is immanent to any living language, whether it takes place through borrowing or coinages that draw lexical material from the native inventory. There is a meaningful distinction, when it comes to endangered and/or minoritized languages, between these two models of lexical expansion. First, active borrowing from the majority language often involves the replacement of traditional lexicon and therefore contributes to language shift (Myers-Scotton, 1992, Romaine, 2010). Second, the coinage of neologisms is associated with language planning activities and purist ideologies, since it is assumed that in bilingual contexts, speakers find it more ‘natural’ to use loanwords from the majority language than to invent new forms based on native resources (Dorian, 1994, Johanson, 2002).

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Nahuatl, the most widely spoken indigenous language of Mesoamerica, has been in a disadvantageous relationship with Spanish for most of their five centuries of co-existence. During the colonial period, which spans from the invasion of Cortés (1519–21) until the Mexican War of Independence (1810–21), Nahuatl managed to maintain a relatively high social status and served as a language of church and legal documentation. After independence, however, pervasive assimilationist policies, particularly in the area of formal education, forced many Nahuatl-speaking communities to abandon their language in favor of Spanish. Nevertheless, there are more than a million and a half speakers of the language today in Mexico (INEGI, 2020), comprising thirty officially recognized varieties spoken across the country (INALI, 2008). And in recent decades there have been multiple grassroots initiatives for the revitalization and revalorization of Nahuatl.

In this study, we deal with a corpus of Nahuatl neologisms, documented in interviews and during specialized workshops involving speakers of five different varieties of the language (see section 3 for more details on the language varieties employed and the methods of data collection). All of the neologisms designate material objects and concepts pertaining to such domains as modern technology, non-endemic fauna, health care and linguistics. They are either 'new', in the sense that they have been used in Nahuatl-speaking communities during the last several decades, or belong to domains that are not commonly discussed in Nahuatl. As far as we know, this is the first multidialectal study of lexical creativity in an Indigenous language of the Americas.

We have chosen to employ the onomasiological theory of word-formation (Štekauer, 2005) for the analysis of our rich data corpus. The onomasiological approach (see section 2 for a discussion of this theory) focuses on the conceptual aspect of naming rather than on the morphosyntactic rules and patterns of word-formation, whose functions in the case of Nahuatl may change diachronically and differ from variety to variety. We also propose to review and enrich the onomasiological approach to word formation through our analysis of the linguistic data from this non-European language.

2. APPROACHING NEOLOGY IN MODERN NAHUATL: IN SEARCH OF A THEORETICAL FRAMEWORK

There have been a number of studies on the phenomenon of 'lexical acculturation' in Indigenous languages of the Americas, beginning from the mid twentieth century: Garvin (1948) for Kutenai, Bright (1952) for Karok, Casagrande (1955) for Comanche, Clark (1977) for Sayula Popoluca, Winter (1992) for Walapai, Verbeeck (1999) for Mopan Maya, and Campbell & Grondona (2012) for Nivacle and Chorote. In these works, lexical acculturation (also sometimes called "linguistic acculturation") has been understood as the adaptation of Indigenous lexicon to the reality of contact with the colonizing culture, a process which also includes the coinage of neologisms with no elements of borrowing.¹

A commonly accepted idea in the studies of linguistic acculturation is the differentiation between 'primary accommodation', based on the use of "native linguistic materials" (Casagrande, 1955: 22) in naming the changing environment, and 'secondary accommodation', which entails the use of loanwords and lexico-semantic calques, thus indicating the stage of mass bilingualism (Casagrande, 1955; Winter 1992; Lockhart 1998). From this perspective, coinages and semantic extensions that do not bear signs of a calque from imported languages are a product of the contact that is cross-cultural but not yet cross-linguistic (involving loanwords and loan translations).

This fundamental difference between coined neologisms and other types of linguistic acculturation seems to be the reason why language contact studies in general have given little attention to Indigenous neology as a phenomenon in its own right, preferring to examine it for the purpose of identifying traces of 'pattern' borrowing in the products of Indigenous lexical creativity (see Zimmermann (2008) and Haspelmath (2009) for the classifications of language contact phenomena). This attitude toward Indigenous neology as merely a stage of lexical acculturation is also responsible for the view that with the intensification of language contact and the subsequent language shift toward the imported language, 'primary accommodation' becomes a more peripheral and therefore less significant phenomenon. Neology in minoritized Indigenous languages then tends to fall out of the scope of linguistics proper, as it is considered a part of 'artificial' human interference in language ecology and thus is assigned as a subject of inquiry for the neighboring and more specialized disciplines of language policy and planning and terminology studies. Terminology studies can indeed provide valuable feedback on the investigation of neologisms. But even within this technically oriented discipline scholars agree that neology is not necessarily a planned activity (Boulanger, 1984; Cabré, 1999: 206–207), and that there is evidence that new terms created by ordinary speakers and not expert planners can gain prominence, becoming an integral part of the daily lexicon (Ameur, 2012).

The studies of Peterson & Webster (2013) and Cowell (2018) show that the propensity for naming things using solely Indigenous lexical resources can continue to operate in languages with a long history of language contact. This tendency is considered the product of a particular language ideology, which encourages speakers to engage in lexical creativity rather than borrowing designations from the dominant code when it comes to naming new objects, concepts and

¹ From the beginning, the studies of acculturation in Indigenous languages mentioned here were an offshoot of a more general framework for the study of acculturation in social sciences, proposed by Redfield et al. (1936).

activities. The process of coining neologisms in such situations cannot be directly associated with lexical acculturation (since new terms are created by people who are already bilingual), nor does it necessarily have a direct connection with or direct implications for language planning. Therefore, a more suitable strategy would be to view and analyze neologisms that arise in a spontaneous or semi-spontaneous manner as products of *lexical creativity*, a more inclusive process, which is manifested in all human languages.

The notion of lexical creativity itself is not unproblematic when it is used to describe word-formation practices. Lipka (2007) maintains that the distinction between *productivity* and *creativity* in word-formation as “rule-governed” vs “rule-changing” (Lyons, 1977: 76-78; Bauer, 1983: 63) is relevant for defining lexical creativity as well. However, in referring to the ‘rule-changing’ property of lexical creativity, authors mainly focus on expressive creativity, i.e. expansion of the lexicon motivated by humorous or poetic intentions and not actual naming (Haspelmath, 2002: 101; Munat, 2015). Our study, on the other hand, mostly focuses on *referential neologisms* (which are both morphosyntactically and semantically novel), one of the four basic categories of lexical innovation according to Lipka (2007: 3). Yet we also consider that the coinage of referential neologisms is an *inherently creative process*, constrained but not strictly determined by *word-formation rules*, since it also implies a combination of social, cultural and individual intellectual factors that influence the origin of new naming units.

In determining the scope of our study, we have chosen to focus on the *lexicogenetic* aspect of lexical creativity and word-formation (Geeraerts, 2002), one that deals with the formation of new naming units and not with speech-play or expressive creativity, although the latter may also manifest itself in our data. This is grounded in the onomasiological theory of word-formation, first conceived by Dokulil (1962) and later developed by Štekauer (1998; 2005). From Štekauer’s perspective, lexical creativity should not be viewed as a phenomenon separate from word-formation productivity, but rather as a complementary-one that usually takes place “within productivity constraints” (2005: 224). The principal argument of onomasiological theory is that each naming unit has an onomasiological structure consisting of a *base*, that reflects the class of a naming unit (Substance, Action, Quality or Circumstance) and a *mark*, that may consist of ‘determiner’ and ‘determining’ constituents (Štekauer, 2005: 215-216). According to the theory, the onomasiological structure of a unit is interrelated with its morphosyntactic structure but does not always coincide with it. In order to establish the relations between the onomasiological components of the unit and morphosyntactic ones, the approach employs the ‘Morpheme-to-Seme-Assignment’ principle, or MSAP. For example, in English the suffix *-or* is one of the means for expressing the onomasiological category of Agent (as in *narrator*), but it also may convey the category of Instrument (as in *generator*), so the correspondence between a morpheme and an onomasiological category is not absolute but context-based (Štekauer, 2005: 216, 218).²

The onomasiological approach always sees the coinage of neologisms as the conscious and creative act of an individual speaker (Štekauer, 2005: 224). From this perspective, the difference between neology as a merely creative process and a terminological or applied linguistic activity is not essential. It is even less relevant in the context of this study, since none of the Nahuatl speakers involved in the creation of neologisms was a professional terminologist; for all of them, neology was a new, unfamiliar activity.

The study of lexical creativity in Nahuatl, an endangered language, must take into account the relations between lexicogenesis and language vitality. Picone (1994) argues that in endangered, “declining” languages, lexicogenesis is characterized by a reduction in word-formation productivity: while certain patterns become increasingly frequent, the traditional word formation inventory as a whole diminishes and no “innovative types” can emerge (1994: 275). While we agree with Picone’s idea that lexicogenesis can constitute a defense mechanism against language decline, our purpose here is to demonstrate that even in the case of a historically marginalized language such as Nahuatl, native speakers who are prompted to engage in lexical creativity can draw on and employ a rich variety of lexicogenetic resources.

3. METHODOLOGY

The analysis presented in this paper is based on three distinct corpora, each with its own characteristics. Their corresponding examples are discussed separately in Sections 4, 5 and 6.

Section 4 presents neologisms excerpted from the transcriptions of linguistic elicitation interviews conducted within the framework of the project “Language as a cure: linguistic vitality as a tool for psychological well-being, health and economic sustainability.” The corpus of interviews is far too extensive and contains far too many examples of neologisms to discuss in its entirety. The analysis of a considerable portion of the terms is presented in Gruda, 2022. In the present paper, we will only consider selected examples relevant to the discussion.

² Semes, according to Štekauer (2005: 215-216), are different from onomasiological categories in that they are more diverse and lack an initial position in the onomasiological structure (base, determined mark or determining mark).

The interviews took place in four different dialectal areas of the Nahuatl language. Each example is provided with a three-letter code identifying the region in which it was collected: GUE for Atlaca in the municipality of Tixtla de Guerrero, Guerrero (Central Guerrero Nahuatl, ISO 639: ngu, 7 interviews); VER for three communities in the municipality of Chicontepec de Tejeda, Veracruz (Eastern Huasteca Nahuatl, ISO 639: nhe, 28 interviews); TLA for four communities in the municipality of Contla de Juan Cuamatzi, Tlaxcala (Tlaxcala-Puebla Central Nahuatl, ISO 639: nhn, 30 interviews); and SLP for the municipality of Matlapa, San Luis Potosí (Western Huasteca Nahuatl, ISO 639: nhw, 20 interviews). The interviewed speakers were of different ages, genders, and educational backgrounds.

The interviews were conducted and transcribed by team members of the research project "Language as a cure": Humberto Iglesias Tepec (Guerrero), Eduardo de la Cruz Cruz (Veracruz), Beatriz Cuahutle Bautista (Tlaxcala), and Elwira Dexter-Sobkowiak (San Luis Potosí). The interviewers are fluent in the respective variants of Nahuatl, and three of them are native speakers. Nahuatl was the first language of the vast majority of the interviewees; in rare cases, when they acknowledged Spanish as their first language, they still spoke Nahuatl fluently and our data comes from the interviews with them that were recorded in Nahuatl. The interviewees were shown first two short video clips and then 31 images; afterwards they were asked to name in Nahuatl the objects and creatures that had appeared in them. Finally, the whole process was repeated: video clips were replayed and photos were shown again, and the interviewees were asked to respond this time in Spanish.

The data presented in section 5 was collected in San Miguel Tenango (Sierra Norte de Puebla) and represents an additional variety of Nahuatl, known as Nahuatl of Western Sierra Norte or Nahuatl of Zacatlán-Ahuacatlán-Tepetzintla (ISO-639 code nhi). Most of the data from the community is a product of terminology workshops conducted within participatory action research designed to promote the introduction of Nahuatl into the work of local health services (Haimovich and Herlinda Márquez Mora, 2020). Five native Nahuatl-speaking members of the community were invited to these workshops in order to produce Nahuatl translations for the Spanish signs already present at the local clinic. The intention was to involve speakers who would differ in age, gender and educational background, so that a minimum of four persons with different characteristics would participate in the creation of each set of terms. Another portion of the data comes from language documentation sessions conducted with community members, which revealed a number of Nahuatl neologisms that had been in use in Tenango for an unknown period of time prior to the beginning of our research. To our knowledge, these neologisms were the product of spontaneous, 'folk' lexical creativity rather than terms created for a specific purpose. All of them designated culturally imported artifacts or concepts.

Section 6 presents Nahuatl terminology for lexical, phonological and morphological analysis as well as punctuation contained in the monolingual dictionary of Modern Huastecan Nahuatl *Tlahtolxitlahucayotl. Chicontepec, Veracruz* (2016). All of the dictionary entries, including these neologisms were created collectively by indigenous college students of the Universidad Autónoma de Zacatecas, who were scholarship recipients of the Instituto de Docencia e Investigación Etnológica de Zacatecas (IDIEZ) between approximately 2002 and 2016.

IDIEZ is a Mexican non-profit corporation that provides a monolingual space for Nahuatl speaking students to participate in teaching, research and revitalization activities related to their native language and culture. All of these students were born and raised monolingually and monoculturally in the traditional, small Nahuatl farming communities of the Chicontepec municipality of the state of Veracruz. (Sullivan, 2011: 139–53).

The process for creating grammatical neologisms consisted first, of collectively discussing a concept, for example, 'prefix', 'inflection', 'vowel' or 'comma', in Nahuatl and from the perspective of the language itself; and second, employing the morphological resources of the language (morphemes and the rules governing their combination) to construct an adequate term which, though not used in everyday speech, would make sense to a native speaker.

All examples in this study are cited using the ACK orthography³, minus the use of the macron for long vowels, adapted to the modern variants of Nahuatl under discussion. The phonetic transcription is provided in the International Phonetic Alphabet. Following Štekauer (1998 and 2005), we capitalize the names of onomasiological categories, such as Object or Number, in order to distinguish them from other uses of the same words.

For the morphological analysis of the examples, we use extensively the grammatical studies of Nahuatl by Andrews (2003) and, to a lesser extent, Launey (2011). This perhaps requires an explanation, since these works deal with the grammar of older, Central variants of Nahuatl and are not concerned with modern varieties of the language. It must be emphasized, however, that what is known as older Nahuatl or, somewhat misleadingly, as "Classical Nahuatl" was not a dialect of Nahuatl on its own, or a codified linguistic standard based on one dialect. On the contrary, it is simply a conventional term for the language of the corpus of written and printed texts from the sixteenth through the nine-

³ The ACK orthography is an enriched, traditional orthography for Nahuatl based on the spelling conventions codified by Horacio Carochi in his grammar of 1645 (2001) and modified by Richard Andrews in his *Introduction to Classical Nahuatl* (2003), Joe Campbell and Frances Karttunen in their *Foundation Course in Nahuatl Grammar* (1989) and Frances Karttunen in her *Analytical Dictionary of Nahuatl* (1992).

teenth centuries, representing—especially in the documents created by native speakers—significant regional, temporal, and idiolectal variation (Olko and Sullivan, 2013: 191–192).

Modern varieties of Nahuatl diverged significantly from their historical ancestors, as all living languages do. This, however, manifests itself mainly in lexis, syntax, and morphosemantics; a significant degree of mutual intelligibility between the variants has been preserved, and speakers of modern Nahuatl have little difficulty understanding older, written texts. Most importantly for the present study, the dialects of Nahuatl discussed in the present paper all draw from the shared repertoire of morphemes described by Andrews and Launey, albeit often employing them with different functions. Although some grammatical studies of the dialects of modern Nahuatl have been published, none of them discusses word formation with as much detail as Andrews. At the same time, we want to make clear that we use Andrews' analysis only as a starting point for our own and, primarily, for labeling morphemes of Nahuatl. When the actual functions of morphemes and their configurations differ from the ones implied in a gloss, they are explained by an additional gloss in square brackets. So when we cite examples of morpheme configurations and their functions from the corpus of older Nahuatl, we are not suggesting that these constitute a genetic ancestor of the Modern examples; rather, we are only illustrating the fact that the functions of morpheme configurations can differ from variant to variant, be they contemporaneous or separated by centuries. Finally, when necessary for clarity, we use curly brackets to delimitate a grouped set of morphemes or words to which an affix is attached as a whole.

4. TERMS FROM THE ELICITATION INTERVIEWS

The nouns *cuahuatl*, 'tree, wood' and *tepoztlī* 'metal' (originally 'copper') normally appear as the left, non-head constituent in Nahuatl compounds, referring to the material composition of the head constituent. *Cua(uh)-* has also been documented since colonial times as denoting different kinds of tools, while *tepoz-* began to be used after European contact to refer to diverse types of metal devices and mechanisms characteristic of European technology. Among the neologisms attested in our study, there are compounds containing these two nouns, and while many involve the position and meanings described above, there are some interesting innovations. Examples include a term for 'doll' (1), which is not necessarily made of wood (the dolls pictured on the elicitation materials are clearly made of plastic), and terms for 'bicycle' (2) and 'car' (3), which are prototypically made of metal. In the latter case, the onomasiological base is, interestingly, *tepoztlī*, which in other denominations of the same object serves as an onomasiological mark (see 6–12).

-
- (1) cuaconetl (TLA)
 k^wa- kone- $\overline{t\dot{t}}$
 wood- child- ABS.SG
 'wooden child' > 'doll'
-

-
- (2) cuachichi (SLP)
 k^wa- $\overline{t\dot{t}it\dot{t}i}$
 wood- dog
 'wooden dog' > 'bicycle'
-

-
- (3) cuatepoztlī (SLP)
 k^wa- tepos- $\overline{t\dot{t}i}$
 wood- machine- ABS.SG
 'wooden machine' > 'car'
-

In some cases, the noun stem *cua(uh)-* has been prefixed to a borrowed noun. It seems that, in the resulting compound nouns, this element is not always semantically salient. The ladder to which (4) refers is made primarily of wood, but the dolls (5) are not.

-
- (4) cuaescalera (VER, TLA)
 k^wa- eskalera
 wood- ladder
 '(wooden) ladder'
-

-
- (5) cuamuñecos (VER)
 k^wa- muñeko- s
 wood- doll- PL
 '(wooden) dolls'
-

The analysis of these examples shows that the semantic relationship between the base noun and the compound created with the mark cua(uh) is varied. Only in (4), the actual meaning of the compound noun can be 'wooden X.' In (1) and (5) it is possible that the nouns were coined once for prototypical dolls made of wood and then used for dolls in general. This usage could have subsequently influenced the pleonastic character of (4). In both cases the function of *cua-* is clearly not to indicate the material from which the referent is made, but rather the class of objects to which it belongs (tools, including toys).

There are over 40 nouns with prefixed *tepoz-* attested in the elicitation interviews from all four communities. It gets attached to simple nouns, both native (6–7) and borrowed (8), as well as to deverbal nouns of different types (9–10).

-
- (6) tepozhuihcol (TLA)
 tepos- wihkol
 metal- hook
 'metal hook'
-

-
- (7) tepozpolocatl (GUE)
 tepos- poloka- $\overline{t\dot{h}}$
 metal- donkey- ABS.SG
 'mechanical donkey' > 'car'
-

-
- (8) tepozcaballo (SLP, TLA, VER)
 tepos- kawajoh
 metal- horse
 'mechanical horse' > 'car' or 'bicycle'
-

-
- (9) tepozmotta (TLA) → computer
 tepos- mo- ta
 metal- RFL- to_see
 'mechanical images' > 'computer'
-

-
- (10) tepoztlamamahquetl (GUE)
 tepos- tlamamahke- tl
 metal- carrier- ABS.SG
 'mechanical carrier' > 'crane'
-

Despite the superficial similarity of all the examples above, their careful analysis reveals important differences. While in (6) the literal and the actual meanings of the term align (e.g. *tepozhuicol* is really a kind of *huicol* 'hook'), in (7) and (10) metaphORIZATION is involved. The function of the element *tepoz-* also seems to be varied: while in (7) and (10–12) it can be translated as 'machine,' there is no such possibility in (6), where it apparently means just 'metal.' Finally, although the vast majority of those coinages refer to things made entirely, or predominantly, of metal, (8) does not.

Both the compounds with *tepoz-* and those with *cua-* are widely attested for the colonial period. The persisting use of this word-formation pattern highlights the remarkable continuity visible in Nahuatl. From the onomasiological point of view, it is clear that the use of the mark for coding the category of Quality in the classification of the numerous terms for tools is original to Nahuatl and lacks any Spanish counterpart.

There are also two terms for 'crane' created by adding the element *tepoz* to nouns which are normally used with an attributive function (11–12). Although it could seem tempting to translate them as 'big machine' and 'heavy machine' respectively, in Nahuatl compound nouns, the final stem is morphologically the head and the preceding stems are modifiers, as can be observed in examples 1–8 (see also Andrews, 2003: 279–282). The terms above should be therefore interpreted as 'mechanical big-thing' and 'mechanical heavy-thing' respectively. Superficially similar examples can be found in older Nahuatl documents; in these, however, the relation between the head and the modifier is different (Launey, 1999), e.g. *cuapitzahuac* should be translated properly as '(he/she is) thin in the head.' In (11–12), on the other hand, an analogous translation (i.e. '(it is) big/heavy in its mechanism') would make little sense.

-
- (11) tepozhuei (TLA)
 tepos- weji
 metal- big
 'big mechanical thing' > 'crane'
-

-
- (12) tepozyectec (TLA)
 tepos- Jektek
 metal- Heavy
 'heavy mechanical thing' > 'crane'
-

Finally, there is also one example in which the noun *tepoztl* has been used as the head of a compound and as an onomasiological base (13). In this case, the term also refers to a machine made of metal:

-
- (13) omeolintepoztl (TLA)
 ome- olin- tepos- tl
 two- to_move[PATN] metal- ABS.SG
 'double-moving machine' > bicycle'
-

*mayi
sunday*

Several interesting lexical items have been used in reference to a picture portraying three figures recognizable as pop-cultural depictions of space aliens. Apart from two loanwords (*zombis* and *ovnis*), they are referred to with Nahuatl words and phrases, presumably created on the spot, as no two speakers used the same denominations. Onomasiologically, these all share a common structure, consisting of a base referring to a class of persons and a mark connoting some specific feature. This feature is expressed in (14) and (15) as the metaphorical quality of being similar to a specific animal ('ant' or 'monkey'), while in (16), it is conveyed as a relation of contiguity.

-
- (14) tzicatlacameh (GUE)
 tsika- t̥taka- meh
 ant- person- ABS.PL
 'ant-people' > 'extraterrestrials'
-

-
- (15) macehualmonohmeh (SLP)
 masewal- monoh- meh
 Indigenous_person- monkey- ABS.PL
 person-like figures > 'extraterrestrials'
-

-
- (16) tlacatl citlalin (TLA)
 t̥taka- t̥t̥ sit̥t̥al- ih
 person- ABS.SG star- ABS.SG
 'star-person' > 'extraterrestrial'
-

(14) and (15) have the traditional structure of a Nahuatl compound noun, in which the mark precedes the onomasiological base. The structure of (16) is innovative, deviating from standard Nahuatl morphosyntactic principles, consisting of the juxtaposition of two absolutive-state nouns, in which the second semantically modifies the first.

(17) is an example of a term that, from the point of view of older Nahuatl morphology as it is described in the standard reference books, would be plainly ungrammatical. In it, the traditional verbal suffix complex *-lo-ni* (Andrews 2003: 345–47; Launey 2011: 165), used to create habitual present non-active forms that then routinely undergo conversion into instrumentive nouns through implicit transposition, is employed as a denominal suffix, or — since adding a denominal suffix to an absolutive-state noun would also be ungrammatical — even as a noun (for a more detailed discussion see [Autor 1, in print]). The mechanism behind this rather startling morphological development is, however, easy to explain in the light of the MSAP principle: since the suffix complex *-lo-ni* was used to represent the onomasiological category of Instrument, the innovation consists in the addition of a word class, that of nouns, to which it can be added, and using juxtaposition rather than suffixation, making it a function word rather than a suffix.

-
- (17) cahuitl loni (GUE)
 kawī- t̥t̥ lo- ni
 time- ABS.SG {NACT- HAB}[INSTR]
 'time tool' > 'clock'
-

There were, however, other intriguing examples, such as when an object that we (as Western researchers) would expect to belong to the category of Instrument, received morphological realizations that in Nahuatl historically correspond to active action and patientive nominalizations (onomasiological categories Process and Patient respectively), as in the case of 'pen' (18, 19). This is despite the fact that the image presented to the speakers depicted only a pen, and not a process or a result of writing.

-
- (18) tlahcuiloltequihuiliztli (GUE)
 t̥t̥ahk^wilol- tekiwi- lis- t̥t̥
 writing- to_use- AAN- ABS.SG
 'use of writing' > 'pen'
-

- (19) tepoztlahcuillo (GUE)
 tepos- t̪ahk^wilo- (l)- -li
 metal- to_write- NACT[PATN]- ABS.SG
 ‘a mechanical written thing’ > ‘pen’

There are also cases of what seems to be phono-semantic matching or, as Blank (2003: 47) characterizes it, word-formation on the basis of “formal similarity” with lexical elements from another language. For example, two persons from San Luis Potosí used the noun *azcahuaztli*, lit. ‘ant-tool’ to signify ‘ladder’ (20).

- (20) azcahuaztli (SLP)
 aska- was- t̪i
 ant- tool- ABS.SG
 ‘ant-tool’ > ‘ladder’

‘Ant-tool’ is neither an obvious motivation for a ‘ladder’, nor typical for Nahuatl names of tools. Noticeably, it shares a considerable amount of phonetic matter with the Spanish *escalera*, especially its first portion: both words begin with a vowel, followed by the consonant group [sk], the vowel [a], and a sonorant consonant. It seems probable that the term started as a loanword from Spanish and then underwent reinterpretation as a Nahuatl compound noun evocative of the humorous comparison of a person climbing a ladder and an ant walking on a vertical surface. The form *azcalera*, used by one speaker in VER, can attest to the intermediate stage of phono-semantic matching.

In a large number of cases, the interviewed persons who were asked to name displayed objects did not reply with one-word labels but with more extensive phrases. Responses in this category range from concise two- or three-word phrases to elaborate descriptions. A limited number of examples consist of nominal phrases that include Spanish prepositions, in this case (21), an amalgamation of the Spanish preposition *de* and the Nahuatl particle *in*. Other multi-word phrases contain subordinate clauses with or without a nominal head (22–24).

- (21) ocuilimeh den atl (VER)
 ok^wili- meh den a- t̪i
 worm- ABS.PL of water- ABS.SG
 ‘water worms’ > ‘octopuses’

- (22) tlamanli pampa ce tlahcuilo (TLA)
 t̪aman- t̪i pampa ce Ø- t̪a- hk^wilo- s
 thing- ABS. for one S.3- O.GEN- write- FUT
 SG
 ‘thing for one to write’ > ‘pen’

- (23) tepoznemilli catli quihuica tetl (SLP)
 tepos- nemi- l- li ka t̪i Ø- ki- wika te- t̪i
 machine- go_forward- NACT[PATN]- ABS.SG that S.3- 3SG- take- stone- ABS.SG
 ‘vehicle that takes stones’ > ‘crane’

- (24) tlen ica tlahcuiloah (VER)
 t̪len i- ka Ø- t̪a- hk^wiloa- h
 that P.3SG- with S.3- O.GEN- write- PL
 ‘that with which they write’ > ‘pen’

The status of such phrases within the lexicon of modern Nahuatl is unclear. At first glance, it may seem that the interviewed persons, not knowing what to call a given object in Nahuatl, resorted to describing it and explaining its purpose or other characteristics. Were this the case, it would be clearly fallacious to interpret them as lexical units in Nahuatl. Several observations do not, however, allow for such an easy dismissal. First, the same speakers who used multi-word phrases, used simple label nouns in other cases. Second, many such phrases, especially the more concise ones, are attested in an identical or nearly identical form in numerous interviews. For example, the phrase *tlen ca tlahcuiloah/tlen ca titlahcuiloah/tlen ica tlahcuiloah* as a denomination of ‘pen,’ have been used by four speakers in VER a total of 13 times. At least some of the multi-word phrases appear, therefore, to be more or less established lexical items. An extremely enlightening example was given by a speaker in SLP for ‘octopuses’ (25):

(25)	catli itztoqueh bajaatitlanmeh (SLP)								
	{katli	Ø-	-itstoke-	h	baha-	a-	ti-	t̄lah}	meh
	{that	S.3-	to_be-	PL	deep-	water-	LIG-	LOC-}	PL
	‘those that are in the deep water’ > ‘octopuses’								

Here there is little doubt that the entire multi-word phrase is treated as a noun by the speaker. The giveaway is the plural *-meh* which obviously does not refer to it either morphologically (as relational words such as *bajaatitlan* cannot be pluralized) or semantically (certainly it is not ‘deep water’ that is pluralized but ‘those who are in it,’ i.e. the ‘octopuses’). It is, therefore, the entire multi-word phrase which is clearly treated as a noun and pluralized. Additionally, the element *-meh* (the default pluralizing nominal morpheme in modern Nahuatl) is here used in a manner which suggests that it is treated as an enclitic rather than a suffix.

5. MEDICAL TERMINOLOGY CREATED DURING THE WORKSHOP IN SAN MIGUEL TENANGO

Regarding the data collected in San Miguel Tenango, it is important to refer first to Nahuatl neologisms that have been in use in the community prior to the participatory action research that dealt with the use of Nahuatl in health care. Some of them can be characterized as neosemes or lexicalizations of elements of the Nahuatl lexicon that were already documented in the early colonial period. For example, *tlamohuizol*, ‘movie’ in Tenango, is a variation of *tlamahuizolli*, ‘miracle’, a patientive nominalization of the verb *mahuizoa*, ‘to be in awe of something’ (Molina, 1880[1571]: 54r, 116r). There are more intricate examples, such as *ixcopinal*, ‘photograph’ (26). It consists of the noun *ix-tli* ‘eye’, and by extension, ‘face’ or ‘surface’ and the patientive noun *copinal*, derived from the verb *-copina*, which means ‘to detach or pull something off’ in a number of modern varieties, including the Nahuatl of Tenango. *Ixcopinal*, literally, is ‘something (an image) detached from a surface’. Compound nouns based on *ix-* and *-copina* that have similar meaning can also be found in other varieties, for example, *ixcopincayotl* in the Huasteca Veracruzana and *ixcopinalli* in the Nahuatl of Montaña Baja de Guerrero (Sullivan, 2016; [redacted], personal communication). This can indicate the presence of a common source for this semantic structure, which managed to spread across the Nahuatl-speaking world in the post-colonial era⁴. In Tenango, the combination of *ix-tli* and *-copina* also serves as the basis for another neologism, *teixcopinani*, ‘photographer’ (27).

(26)	ixcopinal		
	ij-	kopina-	I
	face-	to_copy-	NACT[PATN]
	‘photograph’		

⁴ Molina translates *copina* as *sacar una cosa de otra, o sacar algo con molde*, ‘to extract one thing from another, or produce something with a mold’ (1880[1571], II: 74r). So it is possible that with the formal resemblance of *copina* and the Spanish *copiar* ‘to copy something,’ phono-semantic matching was also involved in labeling ‘photograph’ as *ixcopinal*.

(27)	teixcopinani			
	te-	iʃ-	kopina-	ni
	O.PR-	face-	to_copy-	HAB[AG]
	'photographer'			

Another portion of pre-project neologisms includes compound nouns and nominalizations, such as (28–31).

(28)	pantzaucatl			
	pan-	tsak ^w -	ka-	ṭt
	over-	to_cover-	NPRS [INSTR]- ⁵	ABS.SG
	'umbrella' or 'awning'			

(29)	cuatzaucatl			
	k ^w a-	tsak ^w -	ka-	ṭt
	head-	to_cover-	NPRS[INSTR]-	ABS.SG
	'sombrero'			

(30)	cuapiquiyatl			
	k ^w a-	to_wrap-	ja-	ṭt
	head-	to_wrap-	IMPF[INSTR]-	ABS.SG
	'(slinky) hat' or 'cap'			

(31)	Tenpiquiyatl			
	ten-	piki-	ja-	ṭt
	mouth-	to_wrap-	IMPF[INSTR]-	ABS.SG
	'medical face mask'			

Albeit similar in meaning and structure, examples (29) and (30) bear a significant distinction: while in (30) and (31), imperfect indicative forms are reinterpreted as instrumentive nouns (Andrews, 2003: 345; Launey, 2011: 165), (28) and (29) achieve the same end through the reinterpretation of past tense forms (Andrews, 2003: 352; Launey, 2011: 311); and each of these four absolutive forms end with the singular absolutive suffix.⁶ However, the similarity between (28) and (29) on the one hand and (30) and (31) on the other hand becomes evident in light of the fact that possessive forms of both types of nouns in Tenango share a single structure, i.e. the reinterpreted past tense form of examples (28) and (29), minus the absolutive suffix: nocuatzaucca, 'my sombrero' (32), and nocuapicca, 'my cap' (33).⁷ This particular

⁵ The non-present tense suffix, termed the "participial suffix" by Launey (2011: 69-70), can appear as *-ca*, *-que*, *-qui* or *-c*, and is used in the formation of the preterit, distant past and future tenses, as well as the optative mode and deverbal nouns based on the preterit and the distant past.

⁶ The use of absolutive suffixes with nominalized verb forms is common in some variants of Nahuatl. See Andrews (2003: 325, 340–41) for their use with preterite and customary present agentives in older variants of the language; see Author 3 et. al (2016) for their standard use in the singular absolutive of all preterite agentives in Modern Huastecan Nahuatl.

⁷ Stem modifications in possessive forms vs absolutive forms of deverbal nouns is a known phenomenon in Nahuatl, which can be manifested in omission, addition and even substitution of suffixes (Launey, 2011: 91-93; Andrews, 2003: 345-347, 384-385).

configuration of morphemes, composed of a possessive prefix, a preterite verb root and the non-present suffix, is called an ‘active-action noun’ by Andrews (2003: 351-54) and a ‘noun of state’ by Launey (2011: 311); however, it is important to remember that these names only denote the specific function performed by this configuration in the older variants described by these two grammars. When we contrast Andrews’ and Launey’s descriptions with the data from Tenango, it becomes clear that the correspondence between a morphological configuration and its function may differ from variant to variant. Analysis based on onomasiological theory begins with specific meanings and then identifies the morphological configurations by which they are communicated in a given variant of the language.

(32)	nocuatzauc <u>ca</u>			
	no-	k ^w a-	tsak ^w -	ka
	P.1SG-	head-	to_cover-	NPRS[INSTR]
	‘my sombrero’			

(33)	nocuapic <u>ca</u>			
	no-	k ^w a-	pik-	ka
	P.1SG-	head-	to_wrap-	NPRS[INSTR]
	‘my cap’			

The Nahuatl terms produced during the workshops represent a remarkable diversity of coinage patterns. Among them there are compound nouns, simple agentive nominalizations, compound agentive nominalizations, compound locative nominalizations, noun phrases, imperative clauses and relative clauses. The only example of semantic extension is the agentive *cocoxqui*, which came to mean ‘patient’, whereas morphosemantically it is interpreted as ‘sick person’ with no relation to the person’s clinical status. *Cocoxqui* also gave origin to a new term *cocoxcaittani*, ‘nurse’ (34), which, surprisingly, was proposed by the majority of participants, despite them working individually.

(34)	cocoxcaittani			
	kokoŋka-	ita-	ni	
	sick_person-	see-	HAB[AG]	
	‘one who habitually sees patients’ > ‘nurse’			

A large share of the suggested terminology for Nahuatl signs consisted of multi-word verbal phrases (35–36). Most of these phrases were relative clauses, headed by the relative pronouns *campa* or *canic* (‘where’), or *tlen* (‘what’/‘which’). For example, an elderly female participant showed particular inclination for clauses based on *campa*. Relative clauses were proposed by elderly participants even when a Spanish analogue for the sign was relatively short:

(35)	campa	techpahtiz	in	tepahtani	
	kampa	Ø-	tetŋ-	pahti-	s in tepahtani
	where	S.3-	O.1PL-	to_cure-	FUT DEF doctor
	‘where the doctor will cure us’ > ‘doctor’s office’ (Sp. <i>consultorio</i>)				

(36)	pahtli	tlen	quixnamiqui	cocoliztli	
	pah-	tli	tlen	Ø-	k- iŋnamiki kokolis- tli
	medicine-	ABS.SG	that	S.3-	O.3SG- to_counter illness- ABS.SG
	‘medicine which counters illness’ > ‘vaccines’ (Sp. <i>vacunas</i>)				

Next, there were a number of noun phrases, for example:

(37)	teixpantil ica nipohual
	teijpantil i- ka nipowal
	information P.3SG- with number
	‘information with number(s)’ > ‘statistical information’

(38)	tlahzolcomitl
	tlahsol- komi- tl
	trash- jar- ABS.SG
	‘waste disposal’ (Sp. <i>depósito de basura</i>)

The most remarkable cases, however, arose when participants chose to coin single-root nouns or compound nouns based on traditional word-formation strategies as analogs for Spanish terms. Among these nouns the most numerous category was locative nouns, ending with *-ya-n* (39–41). Some of these locative nouns were based on active verbal stems (39 and 40), while other were based on non-active, or passive verbal stems, i.e. *-lo-ya-n* (41):

(39)	amateilhuiltlalyan
	ama- teilwil- tlal(i)a ja- n
	paper- complaint- put- IMPF[INSTR]- LOC
	‘a place for putting a paper complaint’ > ‘suggestion box’

(40)	tlacolpahtayan
	tlā- koko- l- paht(i)a- ja- n
	O.GEN- be_sick- NACT[PATN]- to_cure- IMPF[INSTR]- LOC
	‘a place for curing injuries’ > ‘treatment unit’ (Sp. <i>curaciones</i>)

(41)	cocoxcehuiloyan
	kokoſ- sewi- lo- ja- n
	sick_person- rest- NACT- IMPF[INSTR]- LOC
	‘a place where patients rest’ > ‘hospital ward’ (Sp. <i>encamados</i>)

Another widespread category of neologisms were habitual agentive nominalizations (42–44), which Andrews labels as ‘customary-present agentive nouns’ (2003: 339):

(42)	tlapalanaltani
	tlā- palana- lta- ni
	O.GEN- to_rot- CAUS- HAB[AG]
	‘something that causes rotting’ > ‘waste receptable’

(43)	tlacehuani
	tlā- sewa- ni
	O.GEN to_extinguish_fire HAB[AG]
	‘fire extinguisher’ (Sp. <i>extintor</i>)

-
- (44) tipoztlaciceyaloni
 tipos- t̥la- siseja- lo- Ni
 metal- O.GEN- to_freeze- {NACT-HAB} [INSTR]
 'a metal object that is an instrument for freezing' > 'cold store' (Sp. *red de frío*)
-

Interestingly, with regard to the last example, other participants suggested a more concise agentive-based term *tlacice-lani*. However, when coined terms for Nahuatl signs were tested with a larger group of speakers, many of them preferred *tipoztlaciceyaloni* over *tlacice-lani* as a more precise designation of *red de frío*.

The instrumentive nouns ending in *-ya-tl*, mentioned above among the 'traditional' neologisms of Tenango, were manifested in the output of the workshops more rarely than the agentive ones, although some examples deserve attention. In (45), the Nahuatl term is actually a loan translation of the Spanish term *punzocortantes*:

-
- (45) tlatzoponi-tlatiquiyameh
 {t̥la- tsoponi- t̥la- tiki} ja- meh
 {O.GEN- to_prick- O.GEN- to_cut} IMPF[INSTR]- ABS.PL
 'instruments that prick and cut > 'sharp objects'
-

In addition, two examples of compound active action nouns (ending on *-liz*⁸) arose when the participants were asked to translate the term 'national insurance' (*Seguro Popular* in Spanish):

-
- (46) altipeyoliliz
 altipe- joli- lis
 city/society- to_flourish- NMLZ
 'flourishing of the society/community' > 'national insurance'
-

-
- (47) pahtlapalehuilizpan
 pah- t̥la- palewi- lis- pan
 medicine- O.GEN- to_protect- NMLZ- LOC
 'under protection with medicine' > '(covered by) national insurance'
-

In the beginning of the workshops, some of the participants, especially the elderly, jokingly complained about the novel and mentally challenging task that had been assigned to them. However, they approached it with all possible responsibility, and it was rare that someone could not suggest any analog in Nahuatl for a particular Spanish term. After two workshops, a poll was conducted among sixty-five speakers from the community, whose first language was Nahuatl, in order to determine their preferences with regard to terminological suggestions for each particular Spanish sign. Then, a final workshop was organized with participants of previous workshops and a few new participants, during which the results of the poll were presented, and, after a discussion, final versions of Nahuatl signs were agreed upon.

6. LINGUISTIC TERMINOLOGY CREATED AT IDIEZ

The indigenous college students who worked on the IDIEZ monolingual dictionary of Modern Huastecan Nahuatl produced a number of grammatical neologisms. These include terminology for phonological, lexical and morphological analysis and punctuation marks.

The phonological terms are created using the two types of "active action nouns" described by Andrews (2003: 351-54, 356-62). They involve four levels of description in a descending sequence: utterance, syllable and phoneme; and the final level, which includes vowel and consonant. The first three are based on the type 2 active action noun *caquilitzli*

⁸ In older Nahuatl, active action nouns end with the absolutive suffix, i.e. *X-liz-tli*, but in some modern varieties, including the Nahuatl spoken in Tenango, the absolutive suffix is often omitted.

‘something audible’ (48–50). The level four words, vowel and consonant, are both constructed using the absolutive form of the type 1 active action noun. First, the intransitive verb *tzilini* ‘to ring’ is transformed into an active action noun, *tzil-incayotl* ‘ringing’, and this in turn receives the compounded *tentli* ‘lip’, referring again to human speech (51). The same process occurs with the transitive verb *tlatzacuilia* ‘to obstruct’: *tentlatzacuilcayotl* ‘consonant’, literally ‘an obstructed sound produced by the human voice’ (52).

(48)	tencaquiliztli				
	ten-	kaki-	lis-		$\widehat{\text{t̥t̥i}}$
	lip-	to_hear-	NMLZ-		ABS.SG
	‘uttered word’				

(49)	achitencakiliztli				
	$\widehat{\text{at̥ji}}$ -	ten-	kaki-	lis-	$\widehat{\text{t̥t̥i}}$
	part-	lip-	to_hear-	NMLZ-	ABS.SG
	‘syllable’				

(50)

pil-

at̥ji-

ten-

kaki-

lis-

tsih

DIM-

part-

lip-

to_hear-

NMLZ-

DIM

'phoneme'

(51)	tentzilincayotl				
	ten-	$\widehat{\text{tsilin}}$ -	ka-	jo-	$\widehat{\text{t̥t̥i}}$
	lip-	to_ring-	NPRS-	pertain[NACT]-	ABS.SG
	‘vowel’				

(52)	tentlatzacuilcayotl				
	ten-	$\widehat{\text{t̥t̥atsak}}^{\text{w}}\text{il}$ -	ka-	jo-	$\widehat{\text{t̥t̥i}}$
	lip-	obstruct-	NPRS-	pertain[NACT]-	ABS.SG
	‘consonant’				

Nahuatl has four lexical categories: nouns, verbs, relational words and particles. Each of the first three represent agglutinating structures, and the neologisms created to name them are what [Andrews \(2003: 356–62\)](#) calls type 2 active action nouns (53–55). The term for the final category, particles, is a compound noun: *piltlahtoltzin*, literally ‘a small word’ (56).

(53)	tlatocaxtiliztli				
	$\widehat{\text{t̥t̥a}}$ -	toka $\widehat{\text{fti}}$ -	lis-		$\widehat{\text{t̥t̥i}}$
	O.GEN-	to_name-	NMLZ-		ABS.SG
	‘noun word’				

(54)	tlachihualiztli			
	tl̥a-	t̥jiwa-	lis-	tl̥i
	O.GEN-	to_do-	NMLZ-	ABS.SG
	'verb word'			

(55)	tlapantiliztli			
	tl̥a-	panti-	lis-	tl̥i
	O.GEN-	to_situate-	NMLZ-	ABS.SG
	'relational word'			

(56)	piitlahtoltzin		
	pil-	tl̥ahtol-	tsih
	DIM-	word-	DIM
	'particle'		

The terms for morphological analysis are mostly “patientive nouns” (Andrews, 2003: 363-80), with the exception of one preterite agentive, *tlamotzquiltihquetl* ‘ligature’ (57). *Tlamotzquiltilli* (58) conveys the idea of the general attachment of an affix and subsumes the remaining three. *Tlanetecholli*, *tlapepecholli*, and *tlapihuillilli* express the strong, medium and light attachment of the derivational affixes, inflectional affixes and clitics, respectively, to the Nahuatl word root. Then, exploiting the fact that Nahuatl employs the parts of the human body to organize space and processes, the following three nouns were employed as compounds in order to convey the ideas of *ad-*, *pre-* and *sub-*, respectively: *ten-tli* ‘lip, and by extension ‘edge’; *tzin-tli* ‘anus’, and by extension ‘beginning’; and *tzon-tli* ‘hair’, and by extension ‘end’. So for example, *tlatzinpihuillilli* ‘preclitic’ (59), *tlatenpepecholli* ‘inflectional affix’ (60), and *tlatzonnetecholli* ‘derivational suffix’ (61).

(57)	tlamotzquiltihquetl			
	tl̥a-	motskiltih-	ke-	tl̥i
	O.GEN-	to_attach-	NPRS[AG]-	ABS.SG
	'ligature'			

(58)	tlamotzquiltilli			
	tl̥a-	motskilti-	(l)-	li
	O.GEN-	to_attach-	NACT[PATN]-	ABS.SG
	'affix'			

(59)

tlatzinpihuillilli

tl̥a- tsin- piwi- li- (l)- li

O.GEN- anus- to_add- APPL- NACT[PATN]- ABS.SG

'proclitic'

(60) tlatenpepecholli
 t̥la- ten- pepet̥jo- (l)- li
 O.GEN- lip- to_stick_on- NACT[PATN]- ABS.SG
 ‘inflectional affix’

(61) tlatzonnetecholli
 t̥la- t̥son- netet̥jo- (l)- li
 O.GEN- hair- to_join- NACT[PATN]- ABS.SG
 ‘derivational suffix’

Terminology for punctuation marks was created through semantic extension, based on the visual characteristics of five concrete nouns (62–67):

(62) ichcayolcuitlapilli
 it̥kajol- k^wit̥lapi(l)- li
 mole- tail- ABS.SG
 ‘mole-and-tail’ > ‘period-and-comma’ > ‘semicolon’

(63) ompaichcayolli
 ompa- it̥kajo(l)- li
 twice- mole- ABS.SG
 ‘double mole’ > ‘double period’ > ‘colon’

(64) ichcayolpilcuatzin
 it̥kajol- pil- kwa- t̥sih
 mole- DIM- wood- DIM
 mole and stick’ > ‘exclamation mark’

(65) tlatlahcopilcuatzin
 t̥lat̥lahko- pil- k^wa- t̥sih
 middle- DIM- wood- DIM
 ‘dash, hyphen’

(66) huihcoltlatzauctli
 wihkol- t̥latsak^w- t̥li
 curved- enclosure- ABS.SG
 ‘parenthesis’

(67) tlatentlatzauctli
 t̥laten- t̥latsak^w- t̥li
 sharp- enclosure- ABS.SG
 ‘square brackets’

Two were constructed by compounding *tzontli* ‘hair’, which can mean ‘top’, or as in the above-mentioned examples of morphological terms, the end of a process (68–69).

(68)	tzoncuittlapilli		
	t̄son-	kwit̄t̄api(l)-	li
	hair-	tail-	ABS.SG
	‘hair tail’ > ‘top comma’ > ‘single quotes’		

(69)	tzoncuittlapiltzitzin		
	t̄son-	kwit̄t̄apil-	t̄si~ t̄sih
	hair-	tail-	PL~ DIM
	‘hair tails’ > ‘top commas’ > ‘double quotes’		

In its earliest recorded state Nahuatl did not use plural suffixes with grammatically inanimate nouns, i.e., plants, rocks, manufactured items, and abstract nouns; this however began to change with the increase of pattern borrowing from Spanish. Another way to show the plurality of inanimates is to use the reduplicated form of the diminutive-reverential root *tzin*. This lexicalized form occurs in the just analyzed word *tzoncuittlapiltzitzin* ‘double quotes’, as well as in the final item in the list of terms for exclamation marks (70):

(70)	ichcayoltzitzin		
	it̄[kajol-	t̄si~	t̄sih
	mole-	PL~	DIM
	‘moles’ > ‘periods’ > ‘ellipses’		

The onomasiological motivation for each of the sets of grammatical terminology created at IDIEZ is quite straightforward, perhaps due to the academic nature of the organization. The phonological terms are conceived of as the Process by which each type of sound is produced, and they are constructed using traditional active-action noun morphology. The same morphology is employed to produce the terms naming the three main kinds of agglutinating lexical categories: noun, verb and relational words. These are also thought of as Process: nouns words name their subject; verb words name the action that their subject performs; relational words situate an entity or an action in a context of space, time or manner. The terms for morphological analysis name morphemes that have been affixed to noun, verb or relational roots with a particular function and position. They are thought of as Patient, and are built with the traditional patientive morphology. The one exception is the term for ‘ligature,’ which is not a patient because it has not undergone affixation. Rather, it is an Instrument for linking morphemes. However, the traditional instrumentive morpheme *-ya* is only employed in a few words in Modern Huastecan Nahuatl and is by no means productive. For this reason, the word for ‘ligature’ is constructed with preterite agentive morphology, which is normally reserved for divine, human and animal agents. And finally, the remaining terminology corresponding to the lexical category of ‘particle’ and punctuation marks, all conceived of as Substance (General), is composed exclusively of compound nouns.

7. COMPARATIVE ONOMASIOLOGICAL ANALYSIS OF THE DATA

The lexical data presented above is extremely diverse both from the perspective of semantics and morphosyntax. We can see how, having been prompted to create neologisms, Nahuatl speakers used multiple methods to designate culturally imported concepts from various semantic domains in their first language. The vast geographical distribution of our data shows that in any community where Nahuatl is a living language, the potential for lexical creativity still exists among speakers, and, if tapped, results in various word-formation patterns that otherwise may remain hidden. The data from sections 3 and 4 demonstrates that different varieties and different speakers of Nahuatl can represent a given concept using different morphosyntactic patterns. So while we rely on the terminology employed by [Andrews \(2003\)](#) and [Launey \(2011\)](#) to label the morphemes involved in the processes of lexical creativity, we do not accept the strict correspondence between them and specific functions that are proposed by the authors.

Rather, we have decided to base our analysis of lexical creativity in Nahuatl on the function-centered onomasiological theory of word-formation (Štekauer, 2005) and show what implications it can have, on the one hand, for our understanding of word-formation in modern Nahuatl and, on the other hand, for onomasiological theory itself.

We have compiled a table (Table A1 in the Appendix section) with representative elements of our data that we have selected in order to highlight its diversity. The data is organized according to the Morpheme-to-Seme Assignment Principle (MSAP) (Štekauer, 2005: 216) and to this end we have identified eight conceptual categories corresponding to the onomasiological base: six of them (Agent, Instrument, Location, Patient, State, and Process) are mentioned in Štekauer (2005), to which we have added the categories of Experiencer (described as a 'case role' in Fillmore, 1968) and Substance (General), which represents cases that could not be related to the other types of Substance (Agent, Instrument, Location, Patient and Experiencer). Experiencer is related to Patient roughly in the same manner as Agent is related to Instrument: both Agent and Experiencer imply the animate nature of the substance. Each conceptual category opens a section of the table: the first column stands for onomasiological types (Štekauer, 2005: 218–219); the second represents onomasiological structures; the third consists of particular examples, separated according to their onomasiological structures; the fourth conveys the basic morphosyntactic structure of a term. It should be mentioned that we do not consider the indefinite object markers *tla-* and *te-* as onomasiological marks since they do not bear any particular semantic content; in most of the cases, they are a part of the determined mark, even if morphologically they are separated from it by another element, i.e. a determining mark.

The MSAP is invaluable in elucidating the conceptual categories that are concealed in Nahuatl neologisms, rather than taking for granted the conceptual categories associated with analogous English or Spanish terms. For example, we can see that in their designations of 'pen' some speakers used morphological constructions that in Nahuatl have traditionally corresponded to Patient (19) or Process (18) and not to Instrument. Does this mean that speakers do not understand the true function of the objects that are shown to them? Very unlikely, as speakers deal with them on a daily basis. Does it mean that the semantic differences between Nahuatl morphological structures that historically corresponded to different conceptual categories are not valid in modern Nahuatl? Improbable, for these patterns continue to function within the traditional lexicon used by speakers today. Another option, which seems the most likely, is that among different speakers and lexical creators, the basic conceptual category of an object can vary and, in general, may differ from the conceptual category given to such an object in European culture and European languages.⁹

A vivid illustration of this argument is the term for *séptico*, which usually designates a small compartment inside a clinic used for the temporary storage of contaminated waste materials. The workshop participants definitely understood the meaning of the term, as it was discussed with them in Nahuatl and Spanish before the coinage attempts took place. A "logical" view (which is probably Eurocentric rather than truly logical) would likely assign *séptico* to the category of Location. However, while one of the terminological suggestions in Nahuatl was indeed constructed as a locative nominalization, namely *tlaihtlacoayan*, 'place where things are corrupted', other options included a habitual agentive nominalization, *tlapalanaltani*, lit. '(an entity) that makes things rot', a patientive noun, *tlamahualli*, 'something contaminated', and a relative clause, *canic ce tlapaca*, 'where one cleans things'. Notably, all four neologisms convey the primary function of *séptico*, but in each case this is accomplished in a completely different manner, and mostly on the basis of different conceptual categories. Moreover, during the stage of testing the coined terminology with a larger group of Nahuatl speakers who did not participate in the workshops, about a half of the respondents chose *tlapalanaltani* as an optimal variant for the sign in Nahuatl, while a smaller number preferred *tlaihtlacoayan* or *canic ce tlapaca*.

It can also be noted that whenever the designated object was more ordinary and easier to grasp, the speakers tended to focus on the onomasiological mark, relegating the onomasiological base to a secondary function. For example, 'pen' is associated with the verb *tlahcuiloa* 'to write' in various morphosyntactic constructions (18, 19, 22, 24), and 'key' is associated with the verb *tlapoa* 'to open' in a similar manner (see examples in Gruda, 2022). But even more remarkable is the persistence of two traditional determining marks in the naming of technological devices - *tepoz-* and *cua-*. Since both *tepoz-* and *cua-* correspond to the onomasiological category of Quality, it is clear that there is a strong tendency among Nahuatl speakers to see technological devices as united by the same one or two qualities. This phenomenon has an obvious relation to the cognitive categorization of objects (see Rosch 1978) and we can assume it is a unique feature of Nahuatl culture and lexical semantics.¹⁰

Thus, our data shows that in the case of Nahuatl, creativity constraints in coining neologisms may differ significantly from those characteristic of English neologisms as shown in Štekauer (2005: 224), where the onomasiological base, with all internal variations, remains the same. When we turn to neologisms that designate more abstract or conventional

⁹ Therefore, in Table 1, when the conceptual category of an object in our view explicitly did not agree with the morphosemantic structure given to that object by a Nahuatl speaker, we chose to include both respective onomasiological categories in the description, as in the case of *tepozmotla*, *tepoztlahcuilolli* and *tlahcuiloltequihuiliztli*.

¹⁰ Concerning the meaning of *cua-* in nominal compounds in Nahuatl see the article by Tomicki (2003).

Table A1

Semasiological and morphological analysis of the examples used in the paper.

Type	Onomasiological structure	Examples	Morphosyntactic structure
Basic category: Agent			
1	Patient- Action - Agent	<i>cocoxca-itta-ni</i> (TEN)	N-V-ni
	Circumstance - Action - Agent	<i>te-ix-copina-ni</i> (TEN)	
Basic category: Experiencer			
1	Experiencer - State- Location	<i>catli - itztoqueh - bajaatitlanmeh</i> (SLP)	catli V N
3(?)	Experiencer - Location	<i>tlacatl - citlalin</i> (TLA)	N + N
		<i>ocuilimeh - den atl</i> (VER)	N de(n) N
Basic category: Instrument			
1	Instrument - Action - Object	<i>tepozmemilli catli - quihuica - tetl</i> (SLP)	N catli V N
		<i>pahtli tlen - quixnamiqui - cocoliztli</i> (TEN)	N tlen V N
	Circumstance - Action - Instrument	<i>cua-tzauc-catl</i> (TEN)	N-V-ca(-tl)
		<i>pan-tzauc-catl</i> (TEN)	
		<i>cua-piqui-yatl</i> (TEN)	N-V-ya(-tl)
		<i>ten-piqui-yatl</i> (TEN)	
	Quality - Process - Instrument	<i>ome-olin-tepoztl</i> (TLA)	N [Num + V > N] + N
	Substance - Action - Instrument	<i>tipoz-tlaciceya-loni</i> (TEN)	N-V-lo-ni
		<i>tepoz-tlamamah-quetl</i> (GUE)	N-V-que-tl
	Object-Action-Process/ Instrument	<i>tlahcuilol-tequihui-liztli</i> (GUE)	-liz(-tli)
	Quality-Action-Patient/ Instrument	<i>tepoz-tlahcuilo-lli</i> (GUE)	-l(-li)
2	Action- Instrument	<i>tlamotzquiltih-quetl</i> (IDIEZ)	V-que-tl
		<i>tlatzoponi-tlatiqui-yameh</i> (TEN)	[V-V]-ya(-tl)
	Instrument -Action	<i>tlamantli pampa - ce tlahcuiloz</i> (TLA)	tlamantli pampa V
		<i>tlen ica - tlahcuiloah</i> (VER)	tlen ica V
	Quality-Action/ Instrument	<i>tepoz-motta</i> (TLA)	N + V = N
3	Object - [Action] - Instrument	<i>tlahzol-[...] - comitl</i> (TEN)	N + N
	Object - [Action] - Instrument	<i>cahuil - [...] - loni</i> (GUE)	N + loni
non-Action	Quality - Instrument	<i>azca-huaztli</i> (SLP)	N + N
		<i>cua-escalera</i> (VER, TLA)	
		<i>cua-tepoztl</i> (SLP)	
		<i>tepoz-huihcol</i> (TLA)	
	Instrument - Quality (?)	<i>tepoz-huei</i> (TLA)	N + adjective-like N
		<i>tepoz-yectec</i> (TLA)	
Basic category: Location			
1	Experiencer - Action - Location	<i>cocox-cehui-loyan</i> (TEN)	N-V-lo-ya-n
	Location - Action - Agent	<i>campa - techpahtiz - in tepahtani</i> (TEN)	campa V N
	Object - Action - Location	<i>tlacocol-pahta-yan</i> (TEN)	N-V-ya-n
	Circumstance/Quality-	<i>ama-teilhuihl-tlala-yan</i> (TEN)	N-N-V-ya-n
	Object-Action- Location		
Basic category: Patient			
1	Circumstance - Action - Patient	<i>ix-copina-l</i> (TEN)	-l(-li)
		<i>tla-ten-pepecho-lli</i> (IDIEZ)	
		<i>tla-tzin-pihuili-lli</i> (IDIEZ)	
		<i>tla-tzon-netecho-lli</i> (IDIEZ)	
2	Action - Patient	<i>tlamotzquilti-lli</i> (IDIEZ)	-l(-li)
Basic category: Process			
1	Circumstance - Action - Process	<i>ten-tlatzacuil-cayotl</i> (IDIEZ)	-ca-yo-tl
		<i>ten-tzilin-cayotl</i> (IDIEZ)	
		<i>altipe-yoli-liz</i> (TEN)	-liz(-tli)
		<i>ten-caqui-liztli</i> (IDIEZ)	
		and its derivations:	
		<i>achitencaquiliztli</i> (IDIEZ)	
		<i>pilachitencaquiliztzin</i> (IDIEZ)	
2	Action - Process	<i>tlachihua-liztli</i> (IDIEZ)	-liz(-tli)
		<i>tlapanti-liztli</i> (IDIEZ)	
		<i>tlatocaxti-liztli</i> (IDIEZ)	

Table A1 (continued)

Type	Onomasiological structure	Examples	Morphosyntactic structure
Basic category: State			
1	Instrument - Action - Process - State	<i>pah-tlapalehui-liz-pan</i> (TEN)	-liz(-tli)
Basic category: Substance (general)			
4	Substance - Quality	<i>pil-tlahtol-tzin</i> (IDIEZ)	N + dim
non-Action	Substance - Substance	<i>ichcayol-cuitlapilli</i> (IDIEZ)	N + N
		<i>ichcayol-pilcuatzin</i> (IDIEZ)	
		<i>cua-chichi</i> (SLP)	
		<i>cua-conetl</i> (TLA)	
		<i>cua-muñecos</i> (VER)	
		<i>huihcol-tlatzauctli</i> (IDIEZ)	
		<i>macehual-monohmeh</i> (SLP)	
		<i>tepoz-caballo</i> (SLP, TLA, VER)	
		<i>tepoz-polocatl</i> (GUE)	
		<i>tlaten-tlatzauctli</i> (IDIEZ)	
		<i>tlatlahco-pilcuatzin</i> (IDIEZ)	
		<i>tzica-tlacameh</i> (GUE)	
		<i>tzon-cuitlapilli</i> (IDIEZ)	
		and its derivation:	
		<i>tzon-cuitlapiltzitzin</i> (IDIEZ)	
		<i>ompa-ichcayolli</i> (IDIEZ)	Num + N
Substance - Number	<i>ichcayol-tzitzin</i> (IDIEZ)	N + pl	

concepts, as in the case of linguistic terminology, constraints on lexical creativity become even less relevant. Unfortunately, we do not have a variety of options for Nahuatl terms produced in the IDIEZ working sessions, as they all were coined in group discussions and accepted unanimously by the participants. However, it is notable how almost all of the IDIEZ neologisms were distributed among only three conceptual categories, with a strict correspondence between conceptual and linguistic categories: morphological terms received the onomasiological base of Patient; parts of speech and phonological terms were designated as Process, and most of the terms for punctuation marks were assigned to Substance (General), as they did not reveal any morphological structure that would allow them to be labeled as Patient, State or Process.

In this manner, the use of specific word-formation patterns in Nahuatl reveals the conceptual category of any given neologism. However, due to its interrelatedness, the MSAP works in the opposite direction as well: conceptual categories may also provide information about changes in Nahuatl morphosemantics. For example, the authors of older Nahuatl grammars distinguish between agentive nominalizations based on the habitual present tense suffix *-ni* and instrumentive nominalizations based on *-NACT-ni* (Andrews, 2003: 339-347; Launey, 2011: 161-165). According to this differentiation, agentive forms are primarily applied to animate subjects. Nonetheless, the data from Tenango, which includes neologisms of both types, shows us that habitual agentive nominalization can be regularly used in the designation of inanimate substances, such as *tlacehuani* 'fire extinguisher', *tlapalanaltani* 'receptacle for the disposal of contaminated waste', and *tlapalehuani*, 'national insurance'. On the other hand, we still have the animate *cocoxcaittani*, 'nurse', and *teixcopinani*, 'photographer'. Hence we can assume that the morphological differentiation between Agent and Instrument in modern Tenango Nahuatl is insignificant. Another noteworthy development is the use of constructions that, according to Andrews (2003) and Launey (2011), correspond to State while naming objects of the conceptual category of Instrument (*cuatzaucatl*, 'sombbrero', and *pantzaucatl*, 'umbrella'). Thus, the categorization of onomasiological bases in Table 1 is based not only on our knowledge of Nahuatl morphology but also on the cognitive semantics of the objects designated by speakers.

Next, we would like to call attention to the distribution of Štekauer's onomasiological types (2005: 218-219) in our Nahuatl data. First, we should allude to the dominance of Type 1, which implies the inclusion of a minimum of three categories (determining mark, determined mark, onomasiological base) in one word-form. Such tripartite structures are especially frequent in our data from the Tenango and IDIEZ contexts, where speakers coined new terminology in an organized manner. Type 2, consisting of Action as an onomasiological mark and an onomasiological base, is represented in a significantly smaller number of neologisms. The prevalence of Type 1 in our data may be the result of two factors. The first is ethnolinguistic: noun incorporation has historically been one of the most salient features of Nahuatl grammar. The second is pragmatic: speakers would be expected to strive for maximum clarity regarding the nature and function of the named object. Both factors may actually be complementing each other, indicating that today's Nahuatl speakers can effectively make use of one of the basic features of their language in order to accomplish practical tasks.

The identification of Štekauer's Types 3 and 4 in our Nahuatl data, however, presents a certain problem. This derives partially from the fact that Štekauer's categorization is based on the explicit or implicit presence of Action as a determined mark, while downplaying the possibility of forming compound nouns without an Action element, which are common in Nahuatl and many other languages. Thus, only a small portion of terms in our data, which bear signs of the impact of language contact, can be confidently assigned to Type 3, for example: *tlacatl citlalin* (likely a calque from the Spanish *hombre estrella*), *cahuittl loni* (see the discussion of this example in section 3), and *tlahzolcomittl*. In *tlacatl citlalin*, lit. 'man star', the action implied is living on or arriving from, while *tlacatl* can be characterized as an Agent and *citlalin* as a Location; in *cahuittl loni* the missed Action element corresponds to counting/measuring, while *cahuittl* stands for Patient and *loni* for Instrument; in *tlahzolcomittl*, *comittl*, 'jar', is a Patient with regard to depositing (Action) *tlahzol*, 'waste'.

On the other hand, in our Nahuatl data we also have many examples of compound nouns in which it is impossible to determine an Action element. The loan blends *tepoz-arillo* or *cua-muñecos* may be interpreted as an Instrument and Patient respectively, but no action associating them with these conceptual categories can be determined. And *ompa-ichcayolli*, lit. '[double]-[cotton seed]', the term for colon, can be analyzed as an endocentric compound [Quality + Substance (General)], whereas *ichcayol-cuitlapilli*, 'semicolon', lit. '[cotton seed]-[tail]', is a [Substance + Substance] structure without a clear distinction between head and determiner, or mark and base for that matter. For such non-Action onomasiological structures, Štekauer (2005: 219) seems to only reserve a place within Type 4, but even so, in the description of this type he does not offer any examples of Substance-based structures.

Our analysis of Nahuatl neologisms suggests that there can be various types of onomasiological structures that do not include any reference (overt or covert) to Action, especially in the context of language diversity, i.e. when the study is not limited to the English lexicon. Rather than assign all such structures to a separate onomasiological type, it makes more sense to divide all onomasiological structures into two principal categories: those that have Action as their conceptual core and those that have Substance as their core.¹¹ In our case, the structures conceptualized around Action thus correspond to Štekauer's Types 1, 2, and 3, while non-Action structures are mostly classified as [Quality + Substance] or [Substance + Substance] (with the possibility that *tepozhuei* and *tepozectec* represent a case of [Substance + Quality]).

Finally, we would like to refer to multi-word phrases, which have a broad presence in our data and are mostly represented by relative clauses. The principles of terminology management generally discourage the consideration of multi-word phrases as neologisms (Cabré, 1999: 205). However, if we consider the relevant examples in terms of unconstrained lexical creativity, the perspective promoted by Filatkina (2018) seems to be more suitable here: "Lexicalized phrases and compounds [in the context of name creation] are equally productive constructions that make distinctions between lexicon (compounds) and syntax (phrases) irrelevant for language users" (2018: 22–23). While we cannot be certain that all of the phrases included into our data represent 'lexicalized phrases', we believe that Filatkina's arguments are highly relevant for the case of Nahuatl.

If we accept the view of multi-word phrases as potential neologisms, we can see that they in fact can be classified within conceptual categories and onomasiological types in the same way as nominalized forms. If in the case of compound nouns it is derivational morphemes that help to determine their conceptual category, then in the case of relative clauses this role is fulfilled by relative pronouns: *campa*, 'where' (35), denotes Location; *tlen*, 'what' (36), *tlen (ica)*, '(with) which' (22), and *catli*, 'which' (21, 23, 25), in most cases refer to an Instrument, and in some cases, when the subject is animate, to an Experiencer; instrumentive constructions are also salient in phrases that include the borrowed preposition *para* (24).

Overall, the influence of language contact on our corpus of neologisms is not as pervasive as it is in ordinary modern Nahuatl speech. Certainly, loan blends, such as *macehualmonohmeh* (15) and *cuaescalera* (4), as well as phrases that use Spanish prepositions, such as *ocuilimeh den atl* (21) are all present in our data. Examples of pattern borrowing are also encountered, as for example, *tlacatl citlalin*, which came from a person whose first language was Spanish. The use of relative clauses and noun phrases instead of more synthetic-agglutinative constructions in the creation of neologisms can also be attributed to the language contact situation (Spanish morphosyntax is significantly more analytic than that of Nahuatl).

However, a huge portion of Nahuatl neologisms, especially those created in the specialized workshops, do not contain lexical borrowings or visible calques from Spanish. The creative task delegated to Indigenous speakers of Nahuatl apparently contributed to their adoption of a somewhat purist approach, prodding them to delve deeply in their knowledge of their own language and retrieve words and patterns that they would otherwise seldom use in daily speech. At the same time, since the neologisms examined here are all experimental in nature, they do not play the role of shibboleths, men-

¹¹ We prefer not to use the labels 'Action-based' and 'Substance-based', as they can create confusion with the notion of onomasiological base.

tioned by Hill & Hill (1986: 123), which serve to distinguish purist speech from everyday language and to boost the self-esteem of the person who coins and uses them. Rather, the diversity of forms produced and the lack of constraints met by the participants in their creation are more in line with the notion of 'positive purism', as formulated by Flores Farfán (2009: 120), referring to an inclusive and productive behavior, different from 'negative purism', which is exclusive and restrictive. Of course, we cannot predict if any of the forms described here will eventually be adopted by the speech communities; however, since they are a part of language planning activities, such as in the cases of IDIEZ and Tenango, these coinages may contribute to the wider use of Nahuatl and therefore to its revitalization in local contexts.

8. CONCLUSIONS

The analysis of lexical innovations created by speakers of different varieties of modern Nahuatl reveals a wide array of morphosyntactic and semantic patterns. This evidence contradicts the notion that in an endangered language the inventory of word-formation patterns inevitably diminishes (Picone, 1994). In our case, the opposite is as true for the least endangered varieties in our study, Eastern and Western Huasteca Nahuatl, as it is for the most endangered one, Tlaxcala-Puebla Nahuatl. In addition to patterns widely attested in the early Nahuatl corpus (several of which have probably experienced semantic change), our experiments in lexical creativity have brought to light some unique morphosemantic constructions, demonstrating previously unknown capacities of lexigonesis in modern Nahuatl. This strongly suggests that prolonged asymmetrical contact with Spanish, widespread bilingualism, an ever dwindling speaker base, and even the break in intergenerational transmission experienced in many regions, have not damaged native speakers' capacity for lexical creativity.

When the methods of data collection applied in this study and the revealed potential for lexical creativity among Nahuatl speakers are juxtaposed, it becomes evident that unless a true, pragmatically based demand for Nahuatl neologisms arises, this potential will remain largely dormant. Apparently, the same reasons that contribute to Nahuatl speakers' decision to cease transmission of their mother tongue to their children also discourage them from being creative in their own language and prompt them to rely solely on loanwords from Spanish. Thus, if in ordinary circumstances lexical creativity in Nahuatl does not flourish, it is not due to some irreversible language depletion but rather to pragmatic factors (the use of loanwords in speech diminishes risks of misunderstanding) and historically formed psychological barriers that prevent speakers from perceiving their language as a fully legitimate tool of communication in all spheres of life.

Our application of the onomasiological framework to modern Nahuatl neology has shown that there is a significant degree of variance between speakers in the use of basic conceptual categories while coining terms for particular referents. The onomasiological and morphosyntactic analysis of neologisms based on the MSAP has demonstrated the existence of significant discrepancies between conventional morphosemantic structures attested in older Nahuatl documents and their diverse functions in the language's modern varieties. We can note here, for example, divergent preferences in naming objects that fall into the onomasiological category of Instrument. Both *-ni* and *-quetl* were used to form agentive nouns in Classical Nahuatl, and Andrews characterizes *-quetl* as an archaic form (2003: 324). Today's Tenango variety continues to employ *-ni* for agentives, but it also uses it to name objects falling into the onomasiological category of Instrument, along with *-lo-ni*, which is actually nothing more than an agentive built on an action impersonalized by the non-active suffix *-lo*. And although the Guerrero and Eastern Huasteca varieties continue to use *-quetl* to form agentives, they also employ it for the onomasiological Instrument classification.

Our study has shown, as the example in the previous paragraph illustrates, that many word-formation patterns associated with designating Process, State, Location and Instrument, which have been substantially described in the grammars of Classical Nahuatl, continue to be fully productive today. However, the use of these patterns by the participants of our study exhibits more semantic diversity than the grammars' descriptions. There have been also a number of cases in which speakers of one variety named an object using different nominalization constructions, each one corresponding to a different onomasiological base. Additionally, during the experiments with lexical creativity, new and unexpected morphosyntactic constructions have arisen, which does not necessarily indicate poor language proficiency on the part of the speaker (e.g. *cahuitt loni* and *tepozmotla* were coined by persons whose first and primary language of communication was Nahuatl). In sum, neology sessions with Nahuatl speakers presented us with more diverse data about word-formation in the language than can be obtained from any text-based Nahuatl grammar.

The framework of the five onomasiological categories as presented by Štekauer has proved useful and has offered valuable insights into the subject matter at hand; however, its application to modern Nahuatl neology has also revealed its limitations. It is therefore clear that the onomasiological approach as a tool should be reworked and expanded in order to improve its efficiency for the study of word-formation patterns in various languages. Our study shows that the Action-centered analysis of naming units, as described by Štekauer in the case of English, may be less relevant when dealing with languages that use different morphosemantic patterns of word-formation. The list of onomasiological categories should also be expanded in order to better correspond to the diversity of semes involved in word-formation.

Similar studies could be conducted, for example, on such synthetic and agglutinative languages of general use as Finnish, Hungarian and Turkish, which have a recent history of lexical engineering, in order to test our suggestions regarding the onomasiological approach and thus contribute to its efficiency as a more universal analytical tool. We should also note that the use of metaphorization and metonymization in the creation of new lexical units (see e.g. Blank, 2003) in Nahuatl, a topic not included in the present paper but nevertheless promising for further research, can also benefit from the application of the onomasiological approach.

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10. STATEMENT REGARDING OVERLAP

The paper "Continuity and change in modern Nahuatl word formation" (in print) has been created as a part of the same research project and is also based on the same corpus of interviews conducted in four communities which is discussed in the section 4. of the present paper. Other than that, it is entirely unrelated: it contains a discussion of Nahuatl terms for six artifacts ('bicycle,' 'car,' 'clock,' 'key,' 'pen,' and 'umbrella'), and focuses exclusively on morphology; the onomasiological approach is not employed there. Its main points are: the continuing use of morphological patterns inherited from the Colonial times; conversly, the rise of innovative morphological patterns; the preference of speakers from various communities for using particular patterns.

Data availability

The authors do not have permission to share data.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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APPENDIX

The following table presents the onomasiological and morphological analysis of the examples used in the paper. See Table A1.

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